



Introducing Readysset Shallow Caching

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Agenda



Agenda

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What problems are we trying to solve?

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**What problems are
we trying to solve?**



The problems we want to solve

Challenges in scaling traditional database architectures

Cost & Efficiency

- **Costs scale linearly** as infrastructure grows, making expansion expensive.
- Repeated identical queries waste compute by **re-calculating the same results**.
- Scaling vertically (bigger hardware) yields **diminishing returns** and hits a hard ceiling.

Scaling Mechanics

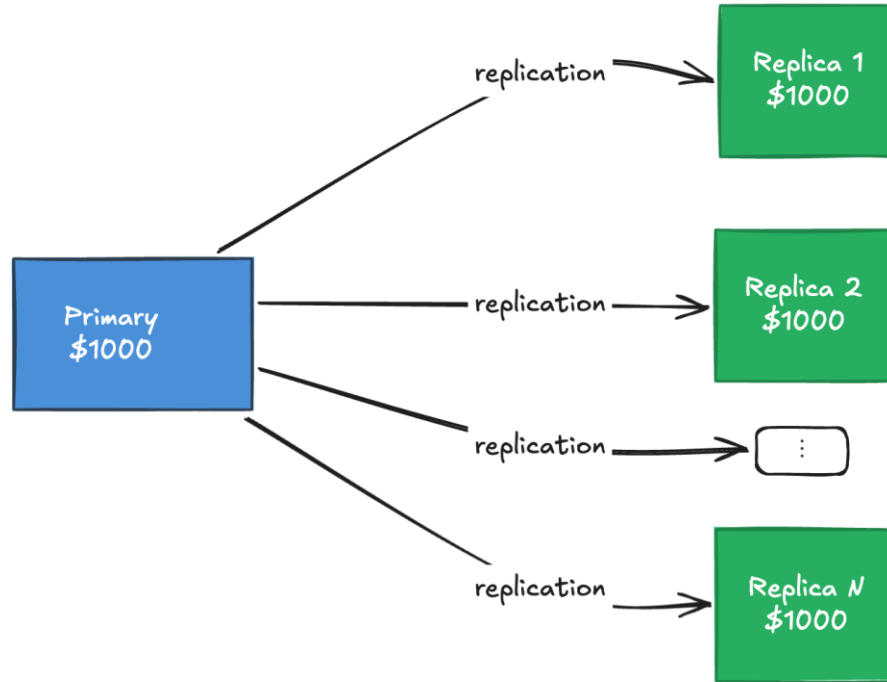
- When database hits max workload, a **new replica** is immediately required.
- Read replicas still **execute the full query**, merely shifting load to a different machine.
- Adding indexes speeds up reads but **slows down writes** significantly.

Complexity Overhead

- Caches (Redis/Valkey) require **rewriting the application** to handle data flow.
- **Cache invalidation** is notoriously hard to get right in distributed systems.
- ORMs often generate **inefficient queries** that are difficult to optimize without code changes.

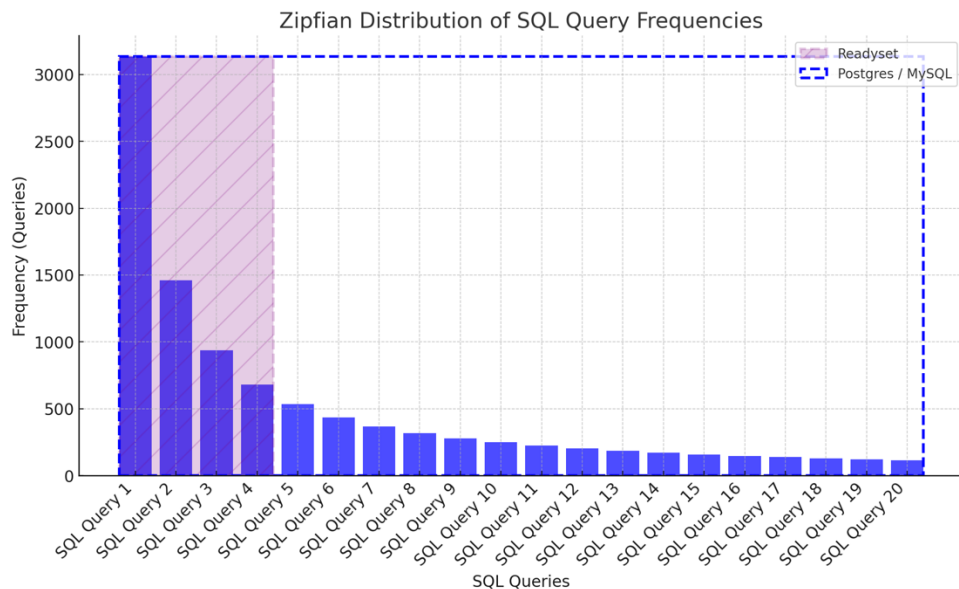


The problems we want to solve



The problems we want to solve

- Queries often follow a Zipfian distribution
- Read heavy workloads. Infrequent changes. Eventual consistency is okay.



Readysset

SQL caching engine



Readysset - A SQL Cache That Just Works

Zero code changes

Readysset speaks native MySQL/PostgreSQL wire protocol and sits transparently between your app and database.

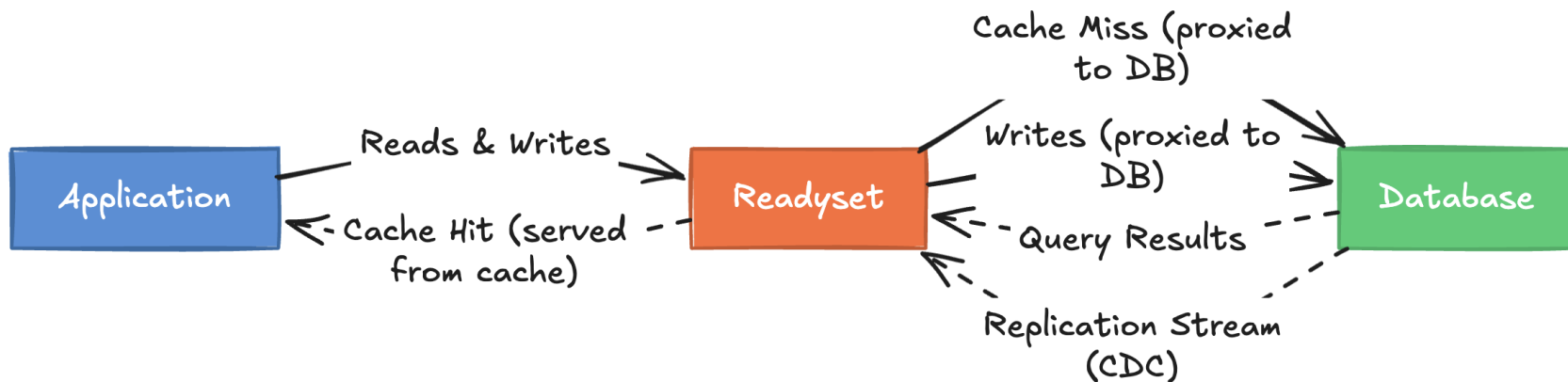
Two caching engines, full SQL coverage

Deep cache uses materialized views kept in sync via replication stream for always-fresh results. Shallow cache uses TTL-based result caching for any query, including complex SQL that deep cache can't handle.

Sub-millisecond reads, less database load

Repeated queries are served from RAM. No manual cache invalidation logic needed.

Readysset – Architecture Overview

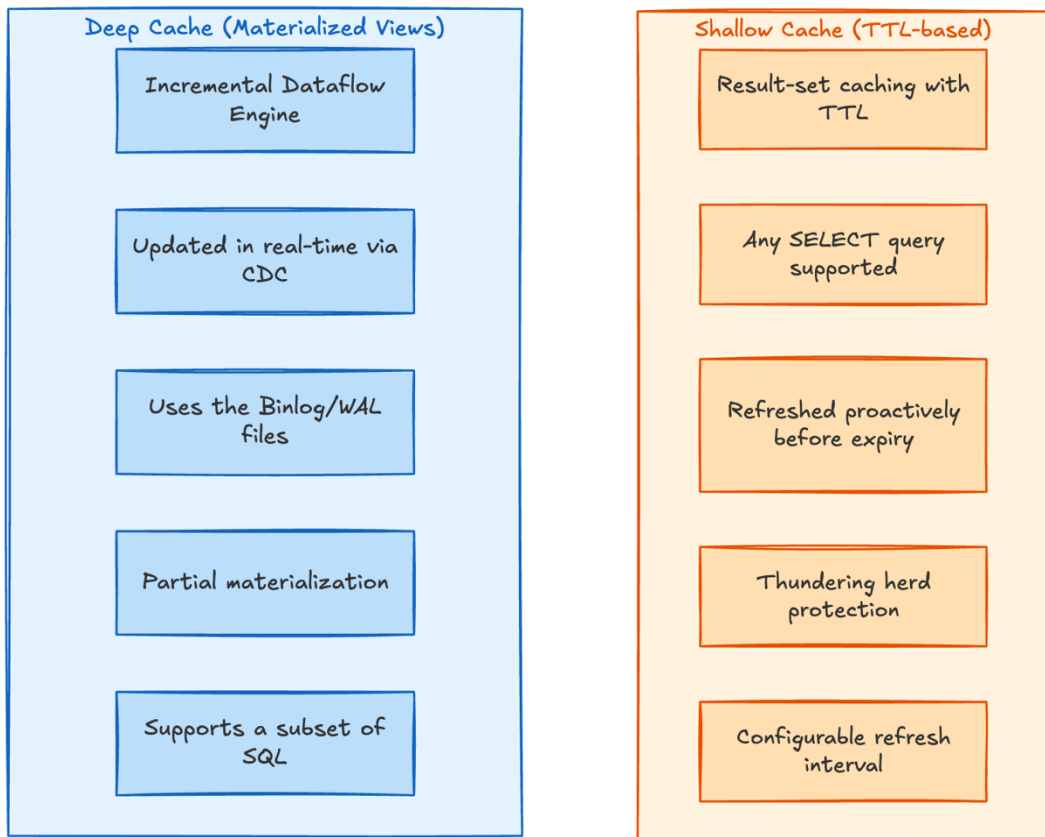


Readysset – Architecture Overview

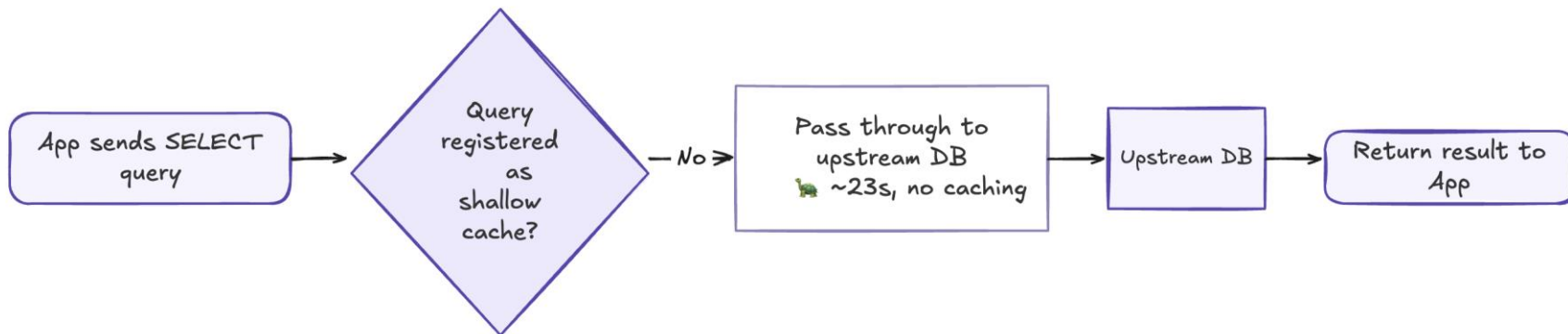
A large orange rectangle with a thin white border, containing the word "Readysset" in a white, handwritten-style font.

Readysset

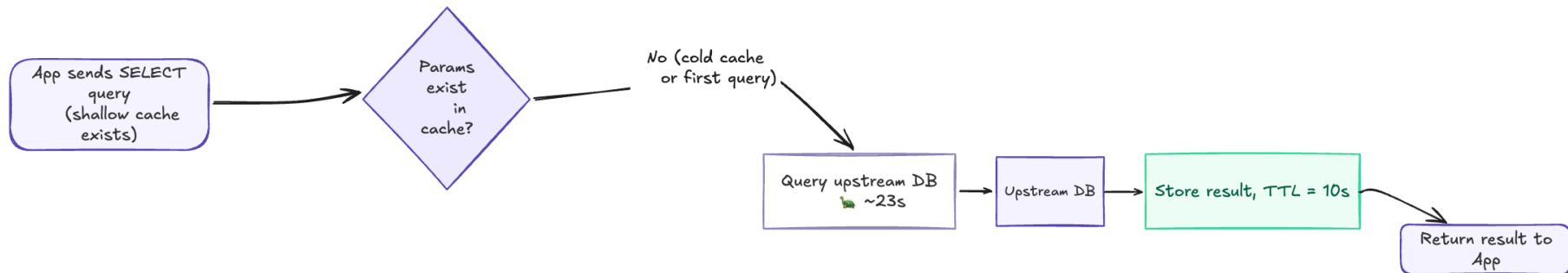
Architecture Overview



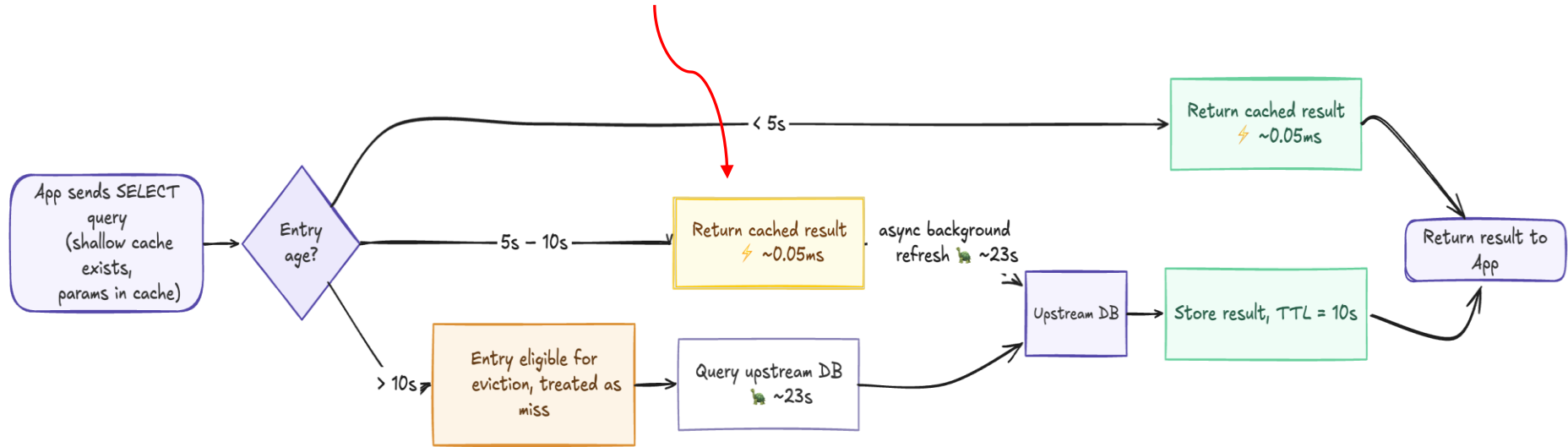
Architecture Overview- Query Not Cached



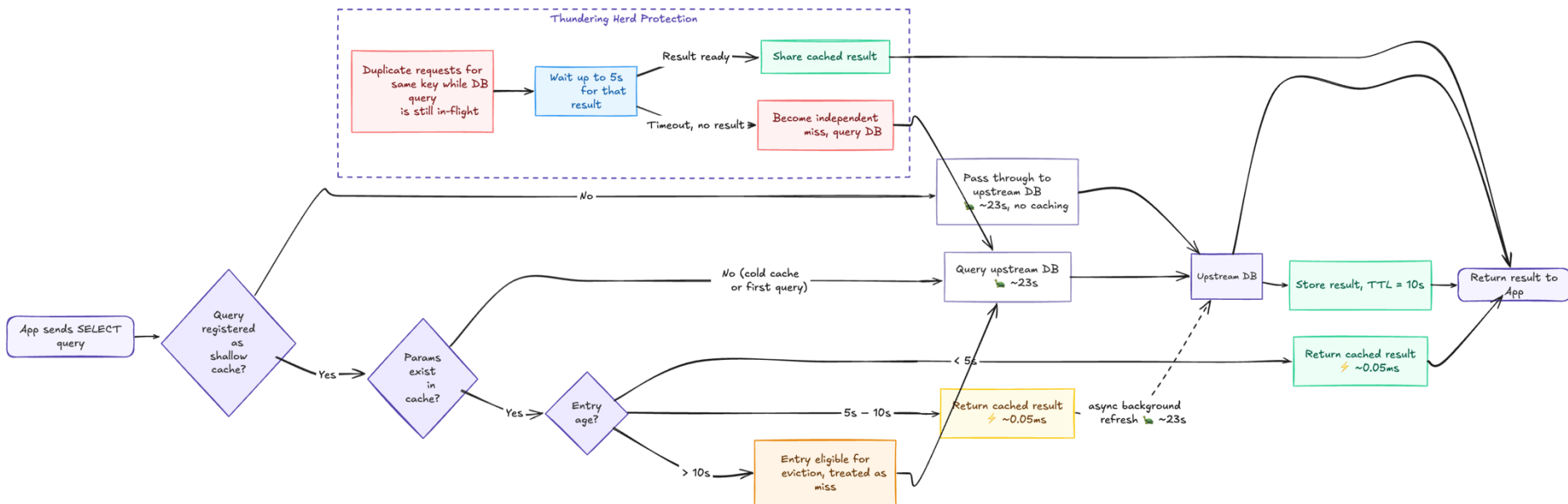
Architecture Overview – Cold Cache



Architecture Overview – Query Cached



Architecture Overview – Shallow Cache Overview



DEMO



Questions?



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```
$ pipx install rdst  
$ rdst configure
```



Thank You!

